R USERS GROUP · INVITED TALK · OCTOBER 2026

Agentic R

Using agentic systems to efficiently write, review and produce reproducible code



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HANDS-ON MATERIALS

github.com/derebellon/agentic-r-workshop

AGENDA

What we will cover today

1. **What agentic AI is**
   From a model that answers to an agent that acts, uses tools and checks itself.

2. **A short history: from autocomplete to agents**
   How coding assistance evolved, and where R fits in.

3. **Agents inside R, and running them locally**
   ellmer, tool calling, and free local models with Ollama.

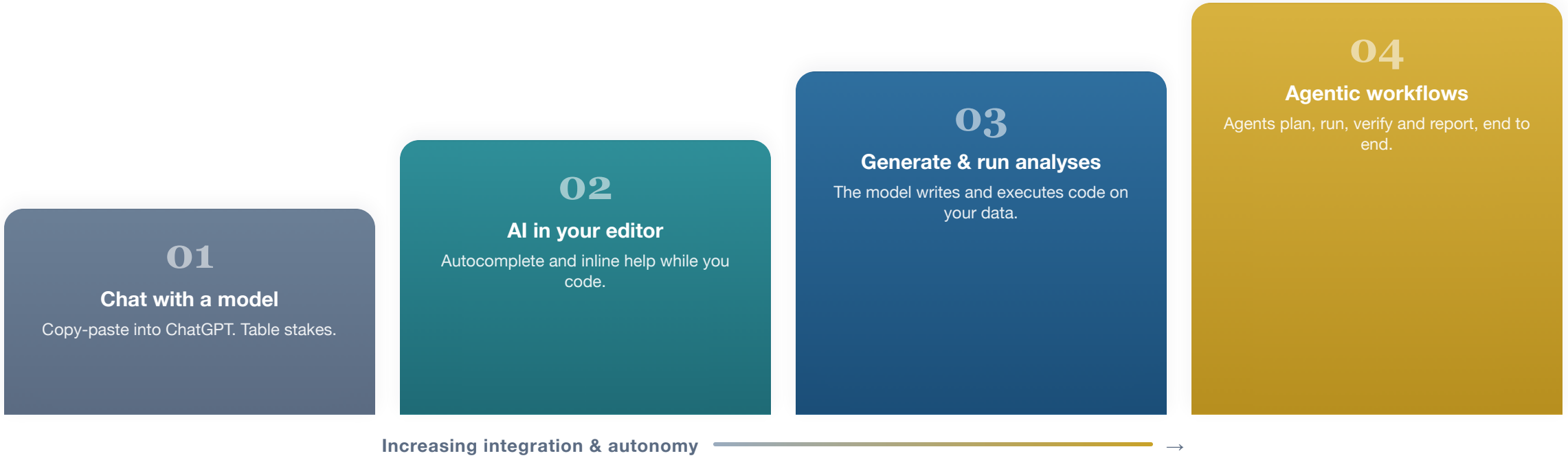
4. **Privacy with sensitive data**
   How to do all of this without your patient data ever leaving your machine.

5. **Live demo**
   An agent that builds a Table 1 and fits a model, on real-looking data.

WHERE MOST OF US REALLY ARE

Four levels of AI maturity

Most people never get past the first step. Each level up means more integration into your real work, and more autonomy.



HOW WE GOT HERE

From autocomplete to coding agents

Help with writing code has climbed the same ladder, from guessing the next word to running whole analyses for you.

~2000S

Autocomplete

The editor suggests the next token or function name. RStudio completion, IntelliSense. It guesses; it does not understand.

→

2021–2023

LLMs generate code

Copilot, ChatGPT. You describe it in words and get a function or a whole script. But it lives outside R, in a chat window.

→

2024–2025

LLMs inside R

Packages like **ellmer** (Posit) let you chat with any model **from R** and give it **tool calling**, so it runs your own R functions.

→

2025 →

Agentic systems

Claude Code and friends: agents that plan, run code, read the error, fix themselves and verify, a reproducible analysis end to end.

> The jump that matters for us: the model stops being a chat on the side and starts **acting inside your R session**, with your data and your functions.

PART 1

What agentic AI is

The shift from a model that answers to an agent that acts, uses tools and verifies itself.

FOUNDATIONS

From language model to agent

FACT

A **language model** (ChatGPT, Claude, Llama, Qwen) is excellent at predicting text: it answers, summarizes, translates and writes code.

HOWEVER

On its own it **does not act**: it does not run your R code, open your files or query your data, and it does not **verify** what it says. Sometimes it **hallucinates**.

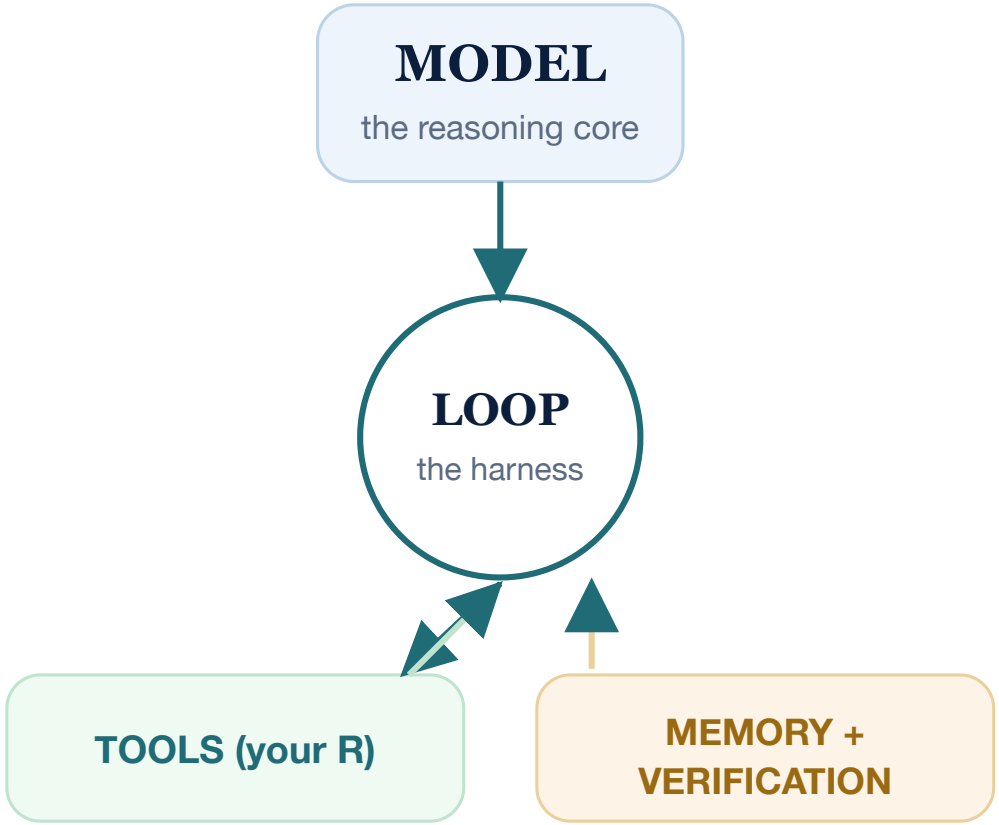
THEREFORE

The **agent** appears: a model + tools + memory + an orchestrator (the "harness") that lets it act, use tools and check itself, with you supervising.

A chatbot **answers**. An **agent does tasks**: it loads your data, runs the model, reads the output, reviews it and reports the result back to you.

FOUNDATIONS

Anatomy of an agent



- **Model**, the reasoning core. Open or closed, large or small, local or cloud.
- **Tools**, run your R code, read files, search literature, query databases.
- **Memory**, knows your project, your methods and your preferences.
- **Verification**, checks its own answers against the data and the output.
- **Human in the loop**, you approve, correct and decide. It accelerates; it does not replace your judgement.

FOUNDATIONS

The harness and the working loop

The **harness** is the software that turns a model into an agent that works in a loop until the task is done: it **perceives**, **reasons**, **uses a tool** (runs code, reads your files), then **verifies and reports**, or repeats.

Where does the harness live?

- **In an environment you own:** your own computer or server, with access only to the files you allow.
- **Sometimes you already use one:** every time you open the **Claude** or **ChatGPT** app, that is a harness running for you.

> In today's workshop the harness is about **50 lines of R**, so you can see every moving part.

```
# the loop, in words
repeat {
  perceive # read your request + state
  reason # decide the next step
  use_tool # run R, read a file
  verify # check the result
  if (done) report()
}
```

FOUNDATIONS

Where can AI run?

You do not always need to send data to a public cloud. Think of it as a spectrum:

More control & privacy ——— **More convenience & power**

Local

On your computer · Ollama

- Runs on your machine
- Data never leaves · free · offline
- Great for short, routine tasks

Cloud

API / provider · Claude, GPT

- Runs in the provider's cloud
- Most powerful, long tasks
- Costs money · check retention

Key message: what matters is **where the model runs** and **what happens to your data**. The same R code works with either.

PART 2

Agents inside R

Chat with any model from R, give it your functions as tools, and run it on a free local model.

AGENTS INSIDE R

LLMs inside R: the ellmer package

ellmer (from Posit) is the bridge: it lets you talk to **any** model from R, local or cloud, with one API.

- **Tool calling:** register your own R functions, the model decides when to run them. That is what turns it into an agent.
- **Structured output:** extract text into a typed table (great for messy clinical notes).
- **Any engine:** a local model with **Ollama** (free, private) or Claude in the cloud, swapped in one line.

> Around ellmer sit the editor assistants: **chattr**, **gander**, **btw**, the agent, right inside RStudio.

```r
# chat with a LOCAL model, free
chat <- chat_ollama("qwen2.5:3b")

# give it one of your R functions
chat$register_tool(tool(
  make_table,
  "Build a Table 1 with gtsummary"
))

chat$chat("Table 1 by treatment")
#> [tool] make_table(by = "treatment")
```

PART 3

Privacy with sensitive data

How to do all of this without your patient data ever leaving your machine.

PRIVACY & COMPLIANCE

How your patient data stays private

Your data stays on your computer. If anything is shared, it is only **de-identified** results.

Your data
Stays on your computer

→

Local analysis
R on your machine

→

Results
Tables, models, figures

→

De-identify
Remove names & IDs

Option 1 · Read locally

A local model (Ollama) interprets the results. **Nothing ever leaves your computer.**

Option 2 · Read in the cloud

Only de-identified results travel. Use a provider under contract that **does not store or train** on your data.

Key message: the sensitive data stays local; only de-identified results may be shared. With a local model, even that stays home.

HANDS-ON

Try it yourself: a practical example

A tiny agent, **"Rho"**, that analyses a synthetic clinical cohort. Everything runs on your laptop.

- **Three folders:** an empty skeleton, the same one filled in, and the agent as a Claude Code project, same idea, three clothes.
- **You ask in plain English:** "Give me Table 1 by treatment, with p-values" and it runs `gtsummary` for real; then "which variables raise mortality?" and it fits a logistic regression.
- **Same code, two engines:** a free local model (Qwen via Ollama) or Claude, switched in one line.

> Clone it, run it, and make it yours: fill in its **soul** and its **tools** with the R libraries you like.

github.com/derebellon/agentic-r-workshop

Thank you

Agents don't replace your judgement, they do the busywork around it.

Model + tools + loop
that is all an agent is

ellmer + Ollama
agents in R, free & private

Local or cloud
same code, your choice

Start small: clone the workshop, run the local agent, give it one of your own R functions, and climb one rung of the ladder. **github.com/derebellon/agentic-r-workshop**